

NTK-Guided Implicit Neural Teaching

Chen Zhang^{*†}, Wei Zuo^{*}, Bingyang Cheng, Yikun Wang, Wei-Bin Kou, Yik-Chung Wu, Ngai Wong
The University of Hong Kong

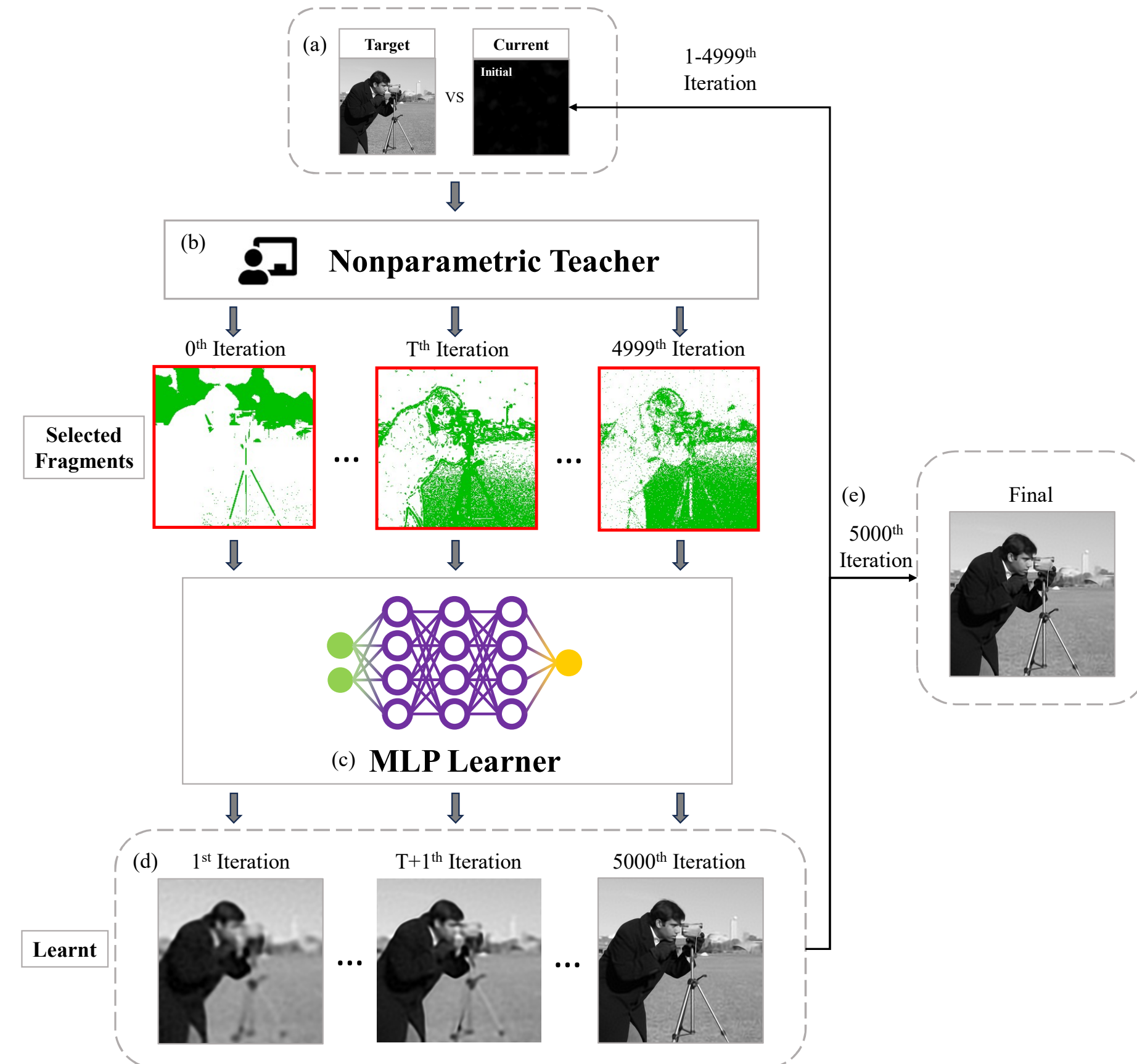


Project Page



Implicit Neural Teaching

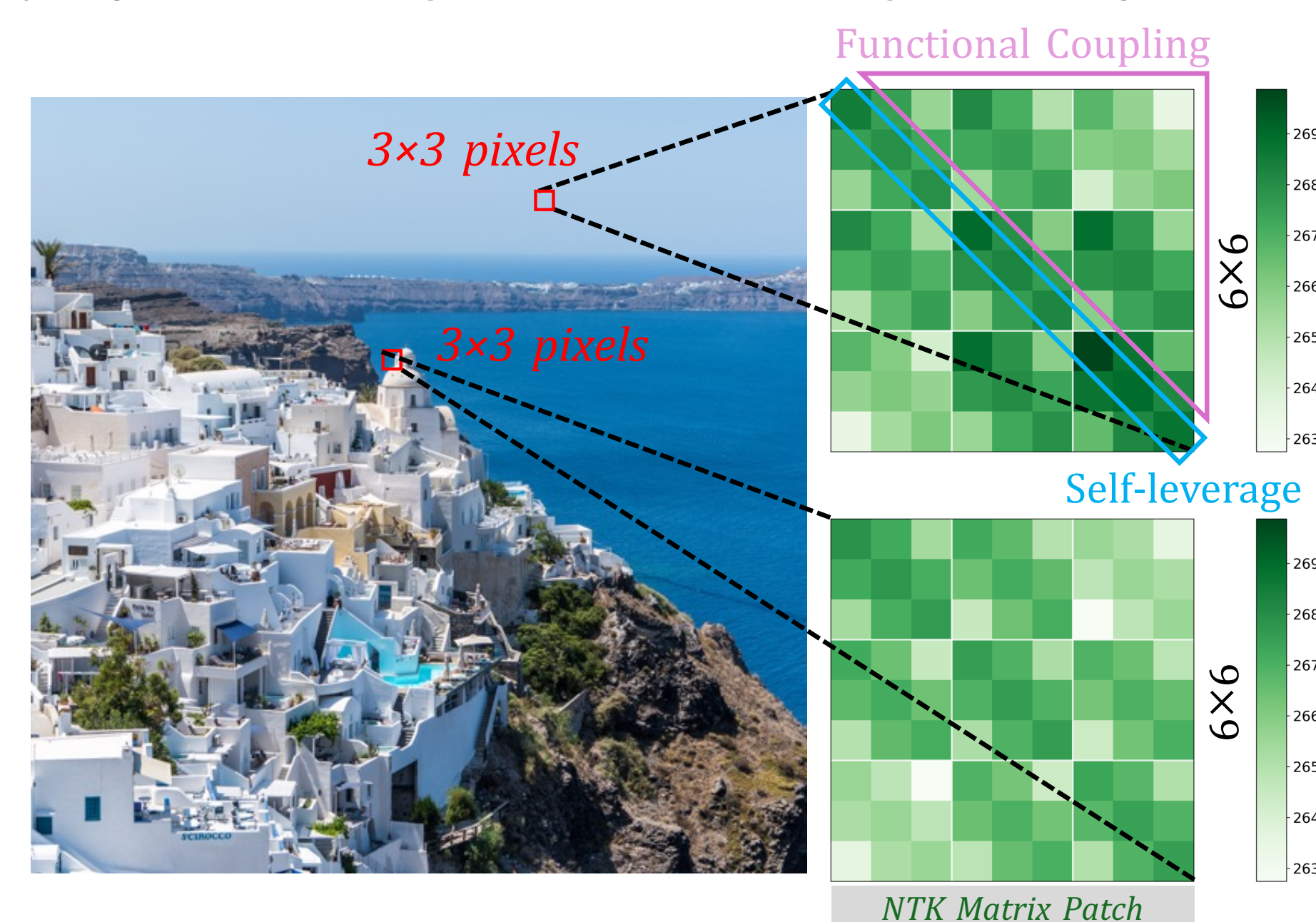
Implicit neural teaching (INT) (Zhang et al., 2024) accelerates implicit neural representation (INR) training by formulating it as a nonparametric teaching problem (*i.e.*, strategically selecting the most informative coordinates to guide MLP convergence, much like a teacher curating high-impact examples), using greedy functional algorithms for coordinate selection (Zhang et al., 2023b,a, 2024, 2025, 2026).



Although effective, it overlooks the MLP's parameter-update dynamics, which limits convergence speed. The Neural Tangent Kernel (NTK) provides a principled way to capture these dynamics. In essence, INT approximates the NTK with a near-identity matrix whose diagonal entries are almost identical. As a result, it cannot accurately track the true update direction unless the NTK is close to diagonal, a condition that rarely holds in typical MLP training.

MLP Dynamics Matter

Instead of relying solely on static output-error metrics, NINT jointly evaluates two complementary factors: (1) the loss gradient with respect to the network output, which identifies regions with large fitting discrepancies, and (2) each coordinate's NTK-induced influence on parameter updates, which quantifies how strongly a point drives global model changes during training.



NINT Algorithm

Algorithm 1 NTK-Guided Implicit Neural Teaching

Input: Signal $S = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$, MLP f_θ , batch size B , learning rate η , iterations T

Output: Trained INR f_{θ_T}

```

1: Initialize  $\theta_0$ 
2: for  $t = 0$  to  $T - 1$  do
3:   Forward pass:  $\hat{\mathbf{y}}_i = f_{\theta_t}(\mathbf{x}_i)$  for all  $i$ 
4:   Compute  $\mathbf{g}^t = [\nabla_{\mathbf{f}} \mathcal{L}(f_{\theta_t}(\mathbf{x}_i), \mathbf{y}_i)]_{i=1}^N$ 
5:   Compute NTK row:  $K_{\theta_t}(\mathbf{x}_i, \cdot)$  for all  $i$ 
6:   Select:  $\mathcal{B}_t = \arg \max_{\substack{\mathcal{B} \subseteq \{1, \dots, N\} \\ |\mathcal{B}| = B}} \| [K_{\theta_t}(\mathbf{x}_i, \cdot) \cdot \mathbf{g}^t]_{i \in \mathcal{B}} \|_2$ 
7:   Update:  $\theta_{t+1} \leftarrow \theta_t - \frac{\eta}{B} \sum_{i \in \mathcal{B}_t} \nabla_{\theta} \mathcal{L}(f_{\theta_t}(\mathbf{x}_i), \mathbf{y}_i)$ 
8: end for
9: return  $f_{\theta_T}$ 

```

Explicitly captures the **heterogeneity in self-leverage** and **functional coupling** encoded in the NTK.

Comparison with State-of-the-art Strategies

Strategy	250 Iterations			1000 Iterations			5000 Iterations		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Stand.	27.897	0.774	0.438	31.669	0.845	0.278	39.763	0.962	0.022
Unif.	27.660	0.771	0.443	31.139	0.836	0.291	37.137	0.943	0.069
EGRA [12]	27.672	0.772	0.443	31.243	0.838	0.289	37.393	0.945	0.068
SoftM. [19]	27.647	0.747	0.498	31.381	0.816	0.338	35.504	0.920	0.070
INT [75]	27.568	0.747	0.499	31.192	0.815	0.351	39.020	0.943	0.035
EVOS [79] (w/o CFS.)	27.964	0.753	0.459	31.657	0.842	0.260	37.175	0.935	0.054
EVOS [79]	28.016	0.755	0.454	31.719	0.842	0.262	37.558	0.940	0.054
Expan. [80] ($\xi = 0.5$) [†]	28.034	0.773	0.420	32.273	0.857	0.227	38.350	0.948	0.049
Expan. [80] ($\xi = 0.7$)	27.990	0.772	0.426	32.154	0.855	0.240	38.220	0.947	0.056
NINT ($\xi = 0.5$)	28.720	0.770	0.438	32.591	0.851	0.246	38.082	0.933	0.037
NINT ($\xi = 0.7$)	28.956	0.776	0.414	32.640	0.841	0.238	39.085	0.958	0.029

[†] denotes the default setting in Expan. [80].

Table 1. Performance metrics (PSNR $\uparrow$, SSIM $\uparrow$, LPIPS $\downarrow$) at fixed training iterations (250, 1000, 5000) across various sampling strategies on image fitting tasks. Purple : the best performance; Pale Purple : the secondary performance.

PSNR	Metric	Stand.	Uni.	EGRA	SoftM.	INT	Strategy EVOS	Expan. ($\xi = 0.5$) [†]	Expan. ($\xi = 0.7$)	NINT ($\xi = 0.5$)	NINT ($\xi = 0.7$)
25	iter $\downarrow$	80	82	82	99	100	132	90	86	72	70
	Time(s) $\downarrow$	8.40	5.94	6.09	7.03	6.44	8.90	6.57	6.34	6.07	5.92
30	iter $\downarrow$	523	626	603	589	589	490	450	454	380	384
	Time(s) $\downarrow$	49.11	38.85	37.85	37.87	33.01	31.20	28.59	29.16	25.63	25.05
35	iter $\downarrow$	2043	2760	2619	2624	2023	2302	1802	1988	1706	1644
	Time(s) $\downarrow$	184.78	168.46	160.41	165.42	111.80	143.20	111.01	123.60	108.90	102.88

[†] denotes the default setting in Expan. [80].

Table 2. Iterations and runtime (in seconds) required to reach target PSNR thresholds across various sampling strategies on image fitting tasks. Purple : the best performance; Pale Purple : the secondary performance.

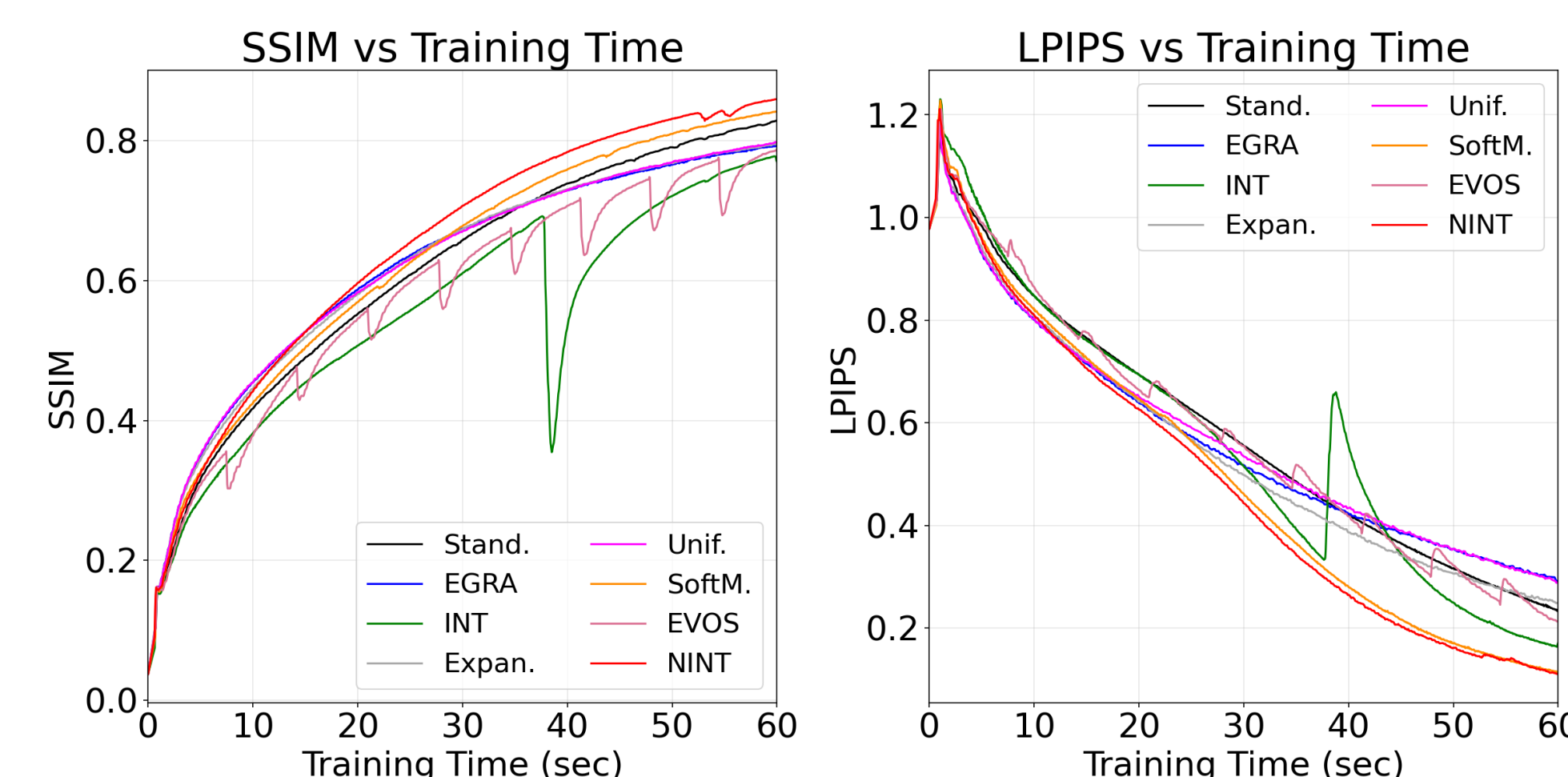


Figure 4. Trends in SSIM [68] and LPIPS [78] metrics over 60 seconds of training across various sampling strategies.

Ablation Studies

Adaptability for Network Sizes

Network Size		3×64	3×128	5×128	3×256	5×256
GT	Stand.	25.22dB	26.08dB	29.40dB	28.14dB	32.67dB
	NINT	25.58dB	26.61dB	30.32dB	29.33dB	34.91dB

Figure 5. Visual comparison of reconstruction quality across various network sizes, after training for 150 seconds on image 07 from DIV2K [1].

Adaptability for Network Structures

Network Structure	PSNR $\uparrow$			Time (s) $\downarrow$
	20s	60s	120s	
MLP	14.91	18.13	20.67	374.64
MLP + NINT	15.11	19.72	21.68	266.32
FFN [65]	16.75	26.9	31.44	54.19
FFN + NINT	18.88	27.39	31.48	48.75
FINER [30]	28.12	31.15	33.01	6.90
FINER + NINT	29.54	31.28	35.61	6.76
GAUSS [48]	21.51	25.87	33.05	56.23
GAUSS + NINT	21.46	27.11	33.38	48.24
PEMLP [65]	18.41	25.09	29.11	59.02
PEMLP + NINT	19.95	25.91	30.45	49.13
SIREN [62]	27.45	30.51	32.44	8.25
SIREN + NINT	29.28	32.40	35.47	5.81
WIRE [55]	16.88	23.86	27.17	83.30
WIRE + NINT	17.62	26.62	29.13	47.23

Table 4. Comparison of PSNR at fixed training times (20s, 60s, 120s) and elapsed time to reach PSNR=25 threshold across various INR network structures, with and without NINT. Better performance is highlighted in purple.

3D Shape Fitting

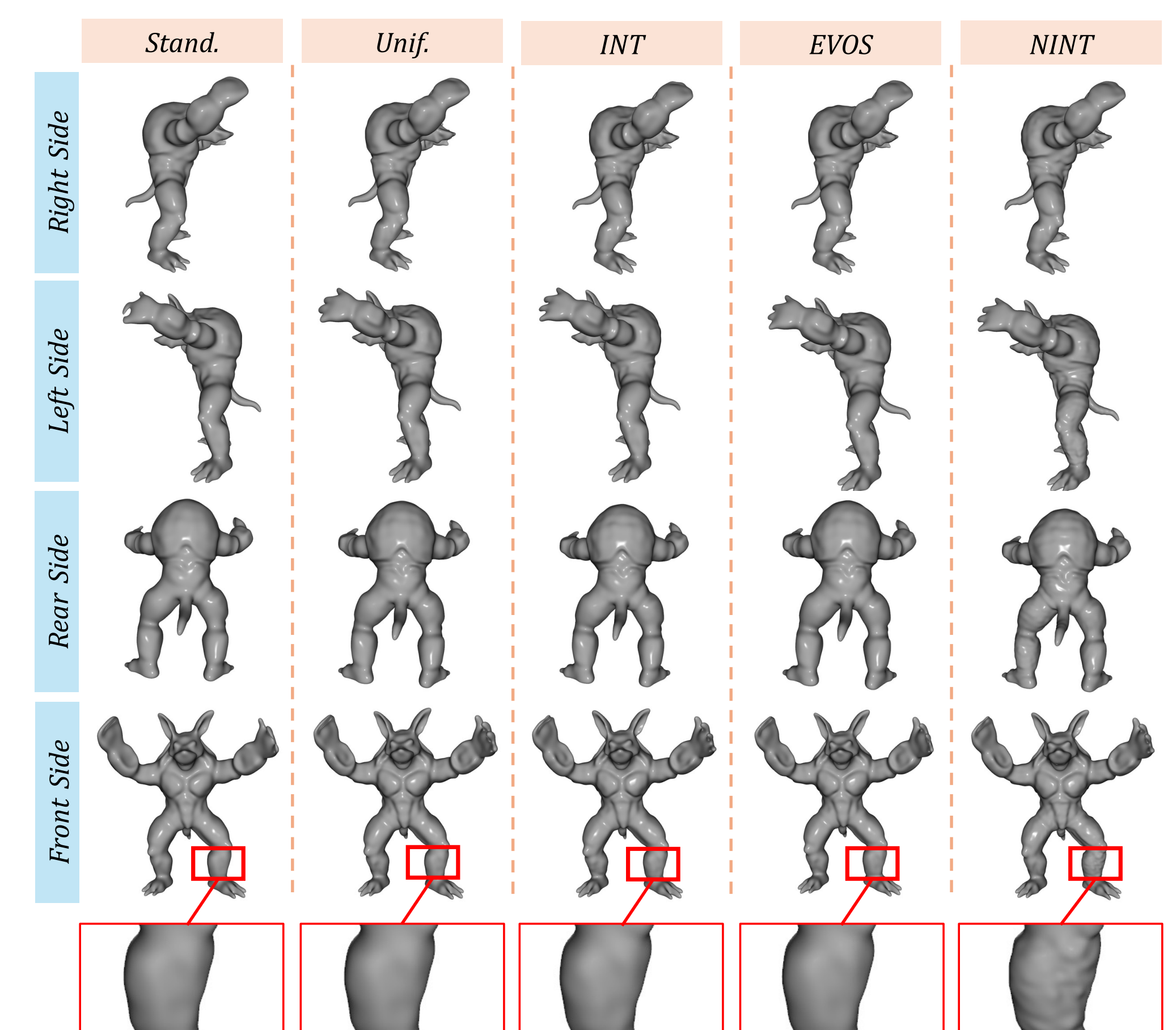


Figure 7. Visual comparison of 3D shape reconstructions using different sampling strategies on the Armadillo of Stanford 3D Scanning Repository [61] dataset, after a fixed training duration of 10 minutes.